

Project Report of CO-OP Project at Industry (Module 2)

On

**Beyond Speed: Evaluating the Cognitive Cost of Accelerated
Learning in Digital Education**

Submitted in partial fulfillment of the requirement for the Award of Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By:

Kirti Sharma	2210990516
Divesh Thapa	2210991528
Harshita Lakhanpal	2210991644

Under Guidance of:

Dr. Isha Kansal



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

Certificate

This is to be certified that the project entitled “**Beyond Speed: Evaluating the Cognitive Cost of Accelerated Learning in Digital Education**” has been submitted for the Bachelor of Engineering in Computer Science & Engineering at Chitkara University, Punjab during the academic semester January 2026 – May 2026 is a Bonafide piece of project work carried out by Kirti Sharma (2210990516), Divesh Thapa (2210991528) and Harshita Lakhanpal (2210991644) towards the partial fulfillment for the award of the course COOP under the guidance of Dr. Isha Kansal.

Sign of Project Guide

(Dr. Isha Kansal)

Candidate's Declaration

I hereby certify that the work which is being presented in the project report entitled **“Beyond Speed: Evaluating the Cognitive Cost of Accelerated Learning in Digital Education”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Isha Kansal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date:

Kirti Sharma
(2210990516)

Divesh Thapa
(2210991528)

Harshita Lakhanpal
(2210990376)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

Dr. Isha Kansal

Department of Computer Science & Engineering,

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Kirti Sharma
(2210990516)

Divesh Thapa
(2210991528)

Harshita Lakhanpal
(2210990376)



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1. Abstract

Digital learning platforms have transformed the modern education system by providing students with flexible educational opportunities and instant access to academic resources. Features such as recorded lectures, playback speed adjustment, online assessments, and digital note-taking have changed the way students consume educational content in online learning environments.

This project evaluates the cognitive impact of accelerated learning in digital education systems. The study focuses on understanding how increased playback speed affects concentration, comprehension, retention capability, note-taking behavior, and mental fatigue among students using online educational platforms.

The research analyzes playback speeds such as 1x, 1.25x, 1.5x, and 2x and compares their impact on learning efficiency and cognitive performance. Survey-based observations, comparative analysis, and behavioral evaluation methods were used to study learner behavior under accelerated learning conditions.

The findings indicate that moderate playback speeds improve productivity and learning efficiency while extremely high playback speeds negatively affect understanding and long-term retention capability.

2. Introduction

2.1 Background

Digital education has become one of the most important components of modern learning systems. Online educational platforms such as YouTube, Coursera, Udemy, and edX provide flexible educational opportunities and allow students to access educational content anytime and anywhere. The rapid growth of online learning systems has significantly changed the way learners consume information and academic content.

Digital education has become one of the most important components of modern learning systems. Online educational platforms such as YouTube, Coursera, Udemy, and edX provide flexible educational opportunities and allow students to access educational content anytime and anywhere. The rapid growth of online learning systems has significantly changed the way learners consume information and academic content. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.2 Accelerated Learning

One of the most widely used features in online learning systems is adjustable playback speed. Students often increase playback speed to complete lectures quickly and improve productivity. Playback speeds such as 1.25x, 1.5x, and 2x are commonly used by learners who aim to save time while covering large amounts of educational material. One of the most widely used features in online learning systems is adjustable playback speed. Students often increase playback speed to complete lectures quickly and improve productivity. Playback speeds such as 1.25x, 1.5x, and 2x are commonly used by learners who aim to save time while covering large amounts of educational material. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.3 Problem Statement

Although accelerated learning improves content consumption speed, its impact on comprehension and retention remains a major concern. Educational videos often contain conceptual explanations, technical diagrams, formulas, and visual information that require

sufficient processing time. Excessive playback speed may reduce understanding and negatively affect learning quality.

Although accelerated learning improves content consumption speed, its impact on comprehension and retention remains a major concern. Educational videos often contain conceptual explanations, technical diagrams, formulas, and visual information that require sufficient processing time. Excessive playback speed may reduce understanding and negatively affect learning quality. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.4 Objectives of the Study

The project focuses on analyzing how playback speed influences concentration, comprehension, note-taking behavior, and learner efficiency. The study also evaluates cognitive load and mental fatigue among students who regularly consume educational content at high playback speeds.

The project focuses on analyzing how playback speed influences concentration, comprehension, note-taking behavior, and learner efficiency. The study also evaluates cognitive load and mental fatigue among students who regularly consume educational content at high playback speeds. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.5 Digital Learning Platforms

Digital learning platforms have improved accessibility and flexibility in education. Students now depend heavily on recorded lectures, online tutorials, and digital educational resources for academic learning. Digital learning platforms have improved accessibility and flexibility in education. Students now depend heavily on recorded lectures, online tutorials, and digital educational resources for academic learning. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.6 Cognitive Load Theory

According to Cognitive Load Theory, the human brain has limited information processing capacity. Excessive information delivery may reduce understanding and negatively affect retention capability.

According to Cognitive Load Theory, the human brain has limited information processing capacity. Excessive information delivery may reduce understanding and negatively affect retention capability. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.7 Previous Research Studies

Several research studies suggest that moderate playback speeds improve productivity while maintaining comprehension levels. However, extremely high playback speeds may increase cognitive overload and reduce retention capability.

Several research studies suggest that moderate playback speeds improve productivity while maintaining comprehension levels. However, extremely high playback speeds may increase cognitive overload and reduce retention capability. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

2.8 Research Gap

Most studies focus on productivity and time efficiency but provide limited discussion on long term cognitive impact and learner fatigue. Most studies focus on productivity and time efficiency but provide limited discussion on long-term cognitive impact and learner fatigue.

3. Methodology

3.1 Research Design

The project follows a mixed-methods research approach involving survey-based analysis, behavioral observation, and comparative evaluation techniques. The project follows a mixed-methods research approach involving survey-based analysis, behavioral observation, and comparative evaluation techniques. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

3.2 Data Collection

Survey forms and learner feedback were used for collecting responses from students who regularly use online learning platforms. The collected data included playback speed preference, concentration level, retention capability, and mental fatigue observations.

Survey forms and learner feedback were used for collecting responses from students who regularly use online learning platforms. The collected data included playback speed preference, concentration level, retention capability, and mental fatigue observations. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

3.3 Comparative Analysis

Different playback speeds including 1x, 1.25x, 1.5x, and 2x were compared to identify differences in comprehension and learning efficiency. Different playback speeds including 1x, 1.25x, 1.5x, and 2x were compared to identify differences in comprehension and learning efficiency. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

3.4 Cognitive Evaluation

The project evaluates cognitive load and information processing capability among learners using accelerated learning techniques. The project evaluates cognitive load and information processing capability among learners using accelerated learning techniques. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

4. Tools and Technologies

4.1 Python

Python was used because of its simplicity, flexibility, and support for analytical applications. The language also provides strong library support for educational and behavioral analysis. Python was used because of its simplicity, flexibility, and support for analytical applications. The language also provides strong library support for educational and behavioral analysis. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions

4.2 Google Forms

Google Forms was used for collecting survey responses and learner observations. Google Forms was used for collecting survey responses and learner observations. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

4.3 Microsoft Excel

Microsoft Excel was used for organizing datasets, graphical analysis, and comparative evaluation. Microsoft Excel was used for organizing datasets, graphical analysis, and comparative evaluation. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

4.4 Online Learning Platforms

Educational platforms such as YouTube, Coursera, Udemy, and edX were analyzed because they provide playback speed adjustment features commonly used by learners. Educational platforms such as YouTube, Coursera, Udemy, and edX were analyzed because they provide playback speed adjustment features commonly used by learners. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

5. Implementation

5.1 Survey Distribution

Online surveys were distributed among students from different educational backgrounds. The survey questions focused on playback speed preference, learning behavior, and concentration level. Online surveys were distributed among students from different educational backgrounds. The survey questions focused on playback speed preference, learning behavior, and concentration level. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

5.2 Data Organization

The collected responses were categorized according to playback speed preference, retention capability, and note-taking behavior. The collected responses were categorized according to playback speed preference, retention capability, and note-taking behavior. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

5.3 Behavioral Analysis

Playback speeds such as 1x, 1.25x, 1.5x, and 2x were compared to understand how learners respond under accelerated educational environments. Playback speeds such as 1x, 1.25x, 1.5x, and 2x were compared to understand how learners respond under accelerated educational environments. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

5.4 Result Interpretation

The final responses were analyzed systematically to identify patterns in learner performance and educational efficiency. The final responses were analyzed systematically to identify patterns in learner performance and educational efficiency. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

6. Results and Discussion

6.1 Playback Speed Preference

The majority of learners preferred playback speeds between 1.25x and 1.5x because these speeds improved productivity while maintaining comprehension levels. The majority of learners preferred playback speeds between 1.25x and 1.5x because these speeds improved productivity while maintaining comprehension levels. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

6.2 Cognitive Load

Students using 2x playback speed reported increased cognitive load and mental fatigue while consuming conceptual educational content. Students using 2x playback speed reported increased cognitive load and mental fatigue while consuming conceptual educational content. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

6.3 Retention Capability

Students learning at moderate playback speeds showed better retention capability compared to learners consuming educational videos at extremely high playback speeds. Students learning at moderate playback speeds showed better retention capability compared to learners consuming educational videos at extremely high playback speeds. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

6.4 Productivity Illusion

Many learners believed they were learning effectively because they completed lectures faster even though their actual understanding and long-term retention were lower. Many learners believed they were learning effectively because they completed lectures faster even though their actual understanding and long-term retention were lower. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

7. Conclusion and Future Scope

7.1 Conclusion

The study demonstrates that playback speed significantly influences concentration, comprehension, retention, and overall learner performance. Moderate playback speeds improve learning efficiency while maintaining understanding and cognitive comfort. The study demonstrates that playback speed significantly influences concentration, comprehension, retention, and overall learner performance. Moderate playback speeds improve learning efficiency while maintaining understanding and cognitive comfort. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

7.2 Future Scope

Future improvements may include AI-based personalized playback speed recommendation systems, adaptive learning environments, long-term cognitive performance analysis, and machine learning-based educational behavior prediction systems. Future improvements may include AI-based personalized playback speed recommendation systems, adaptive learning environments, long-term cognitive performance analysis, and machine learning-based educational behavior prediction systems. Furthermore, the analysis also highlights how learner behavior changes under time pressure and accelerated educational conditions.

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